

Working with Databases

Total Time: 3 Hours

Theme: Database Setup & Relational Design

Dataset: <https://www.kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurants>

Section A: Concept Application

1. You are setting up a new database for the Zomato project. Which SQL command creates a new database called `analytics_db`?
2. You add a `restaurant_id` column to the `restaurants` table as its primary key. What is a primary key, and why must its values be unique?
3. The `ratings` table references `restaurant_id` from the `restaurants` table. What is a foreign key, and what integrity rule does it enforce?
4. You apply a NOT NULL constraint to the `restaurant_name` column. What kind of data entry does this constraint prevent?
5. You draw an ER diagram before writing any SQL. What does an ER diagram represent in relational database design?

Section B: Practical Task

1. Develop a Python-based ETL script using `SQLAlchemy` to automate the creation of the `analytics_db` schema and enforce composite primary keys on the `ratings` table.
2. Implement SQL `ALTER` statements to establish Referential Integrity through Foreign Key constraints between `restaurants` and `locations` with `ON DELETE CASCADE` actions.
3. Write a Python validation script using `Pandas` to clean the Zomato dataset, handling "NEW" or "-" values in the rating column before relational ingestion.
4. Construct an optimized SQL indexing strategy on the `location_id` and `cuisines` columns to accelerate multi-table JOIN operations for analytical queries.

Section C: Mini Project

1. **Title:** Predictive Demand Schema Architecture for Food-Tech Analytics
2. **Problem Statement:** Design a normalized, high-performance relational database architecture optimized for real-time feature extraction for restaurant success prediction models.
3. **Dataset Recommendation:** Zomato Bangalore Restaurants Dataset (Kaggle)
<https://www.kaggle.com/datasets/himanshupoddar/zomato-bangalore-restaurants>
4. **Required Deliverables:** Optimized SQL DDL script (.sql), Python-based data ingestion pipeline, ER Diagram (3NF), and a SQL-based feature engineering report.